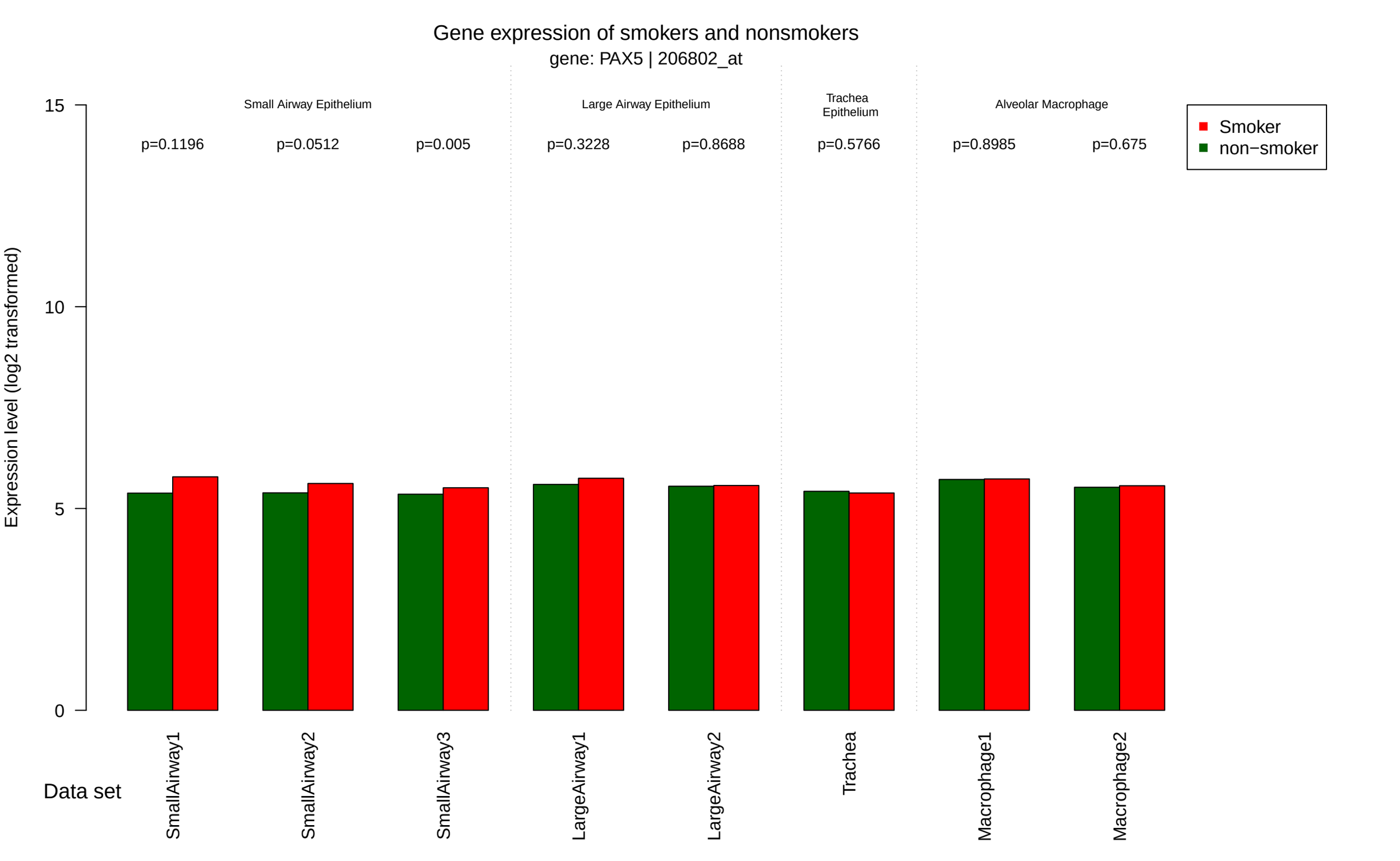
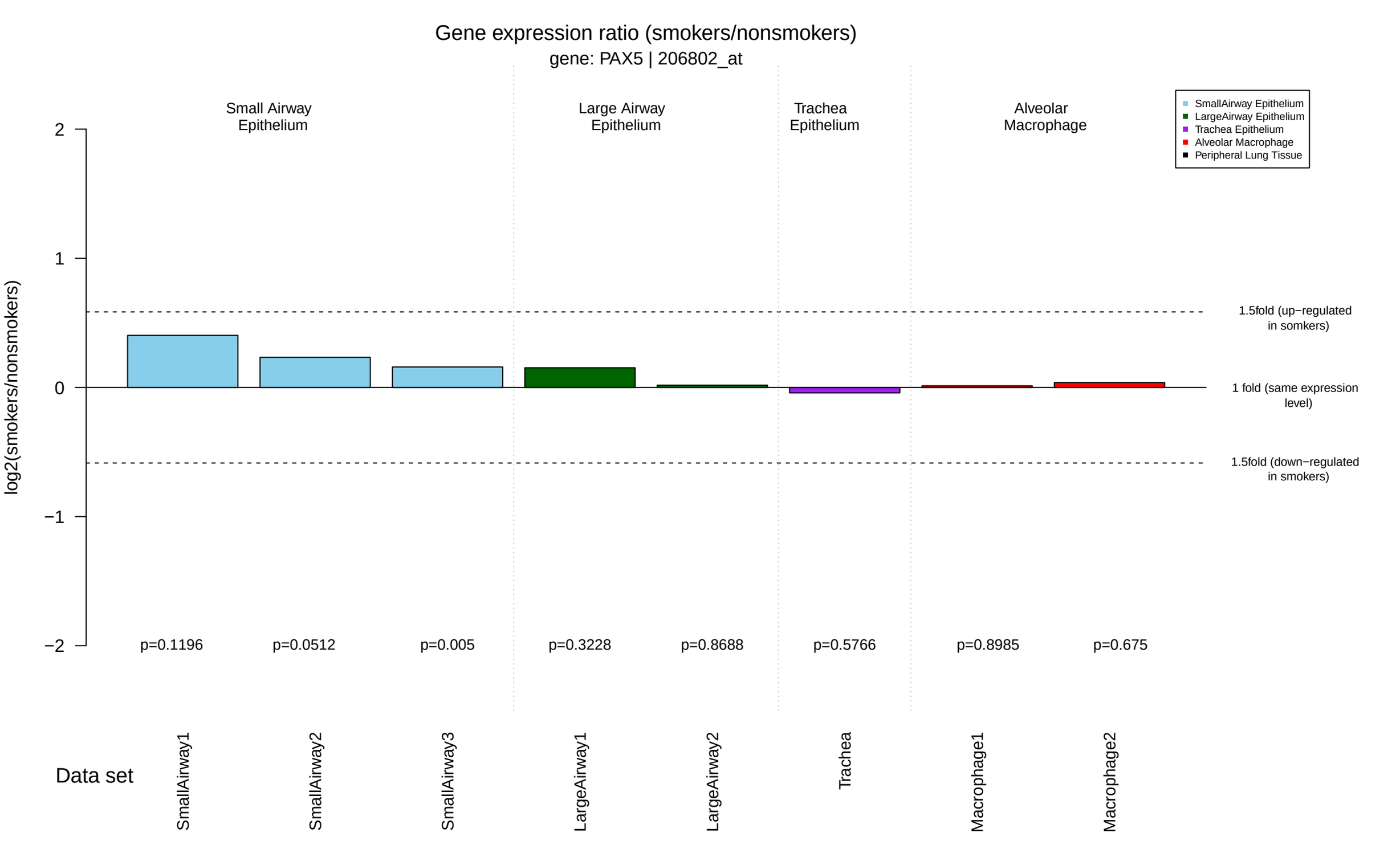
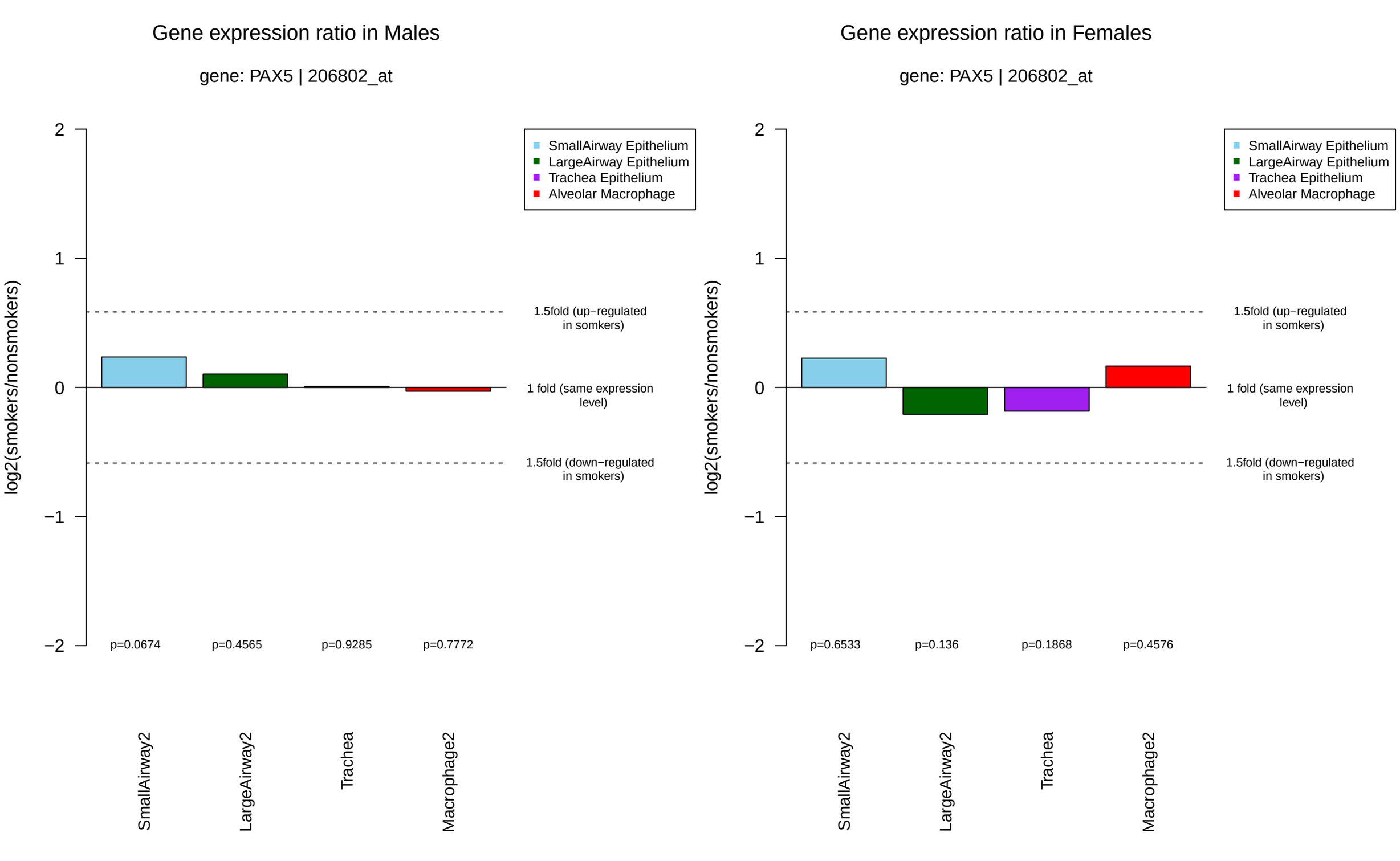
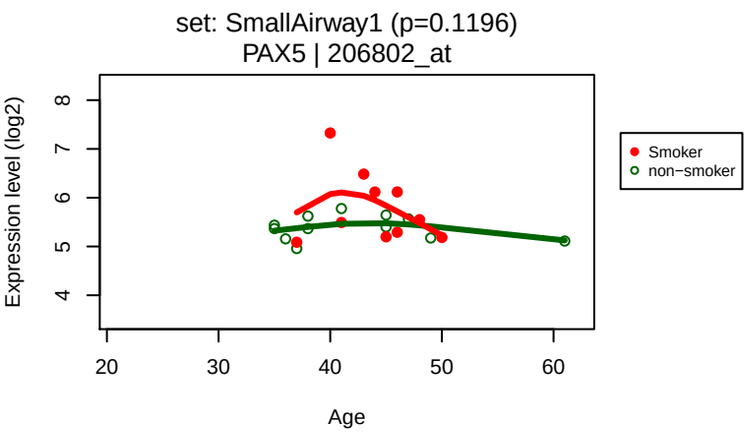
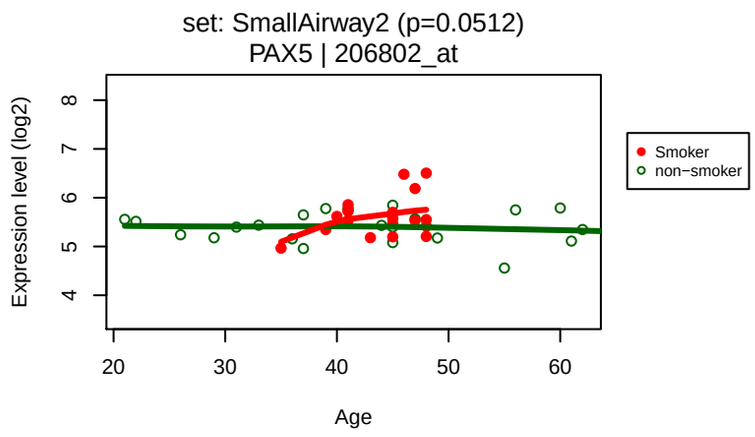
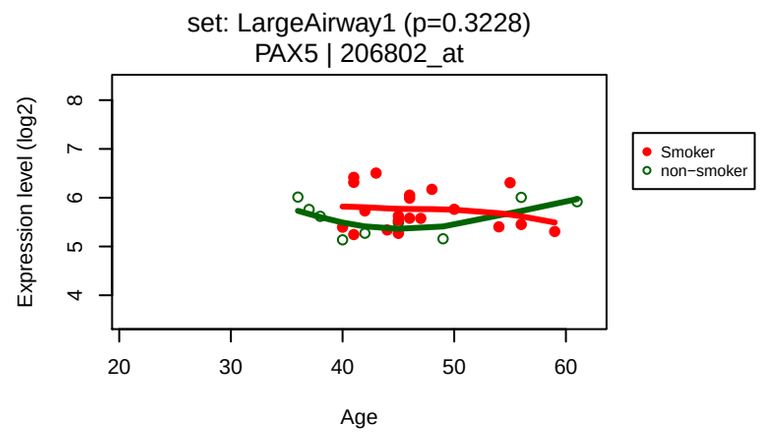

Gene expression of smokers and nonsmokers
gene: PAX5 | 206802_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1196
p=0.0512
p=0.005
p=0.3228
p=0.8688
p=0.5766
p=0.8985
p=0.675
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

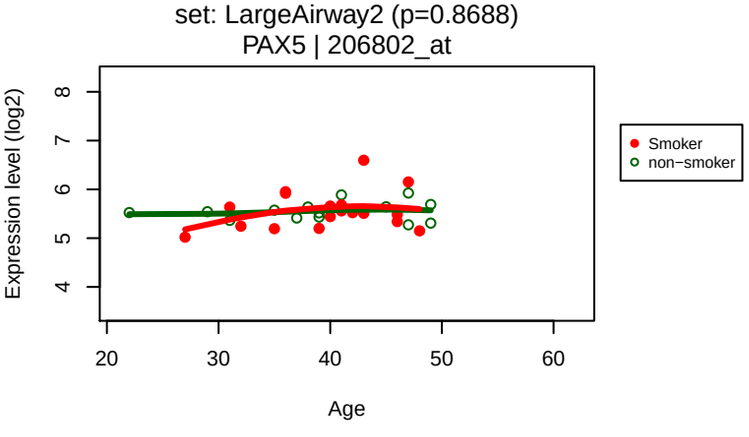

Gene expression ratio (smokers/nonsmokers)
gene: PAX5 | 206802_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1196
p=0.0512
p=0.005
p=0.3228
p=0.8688
p=0.5766
p=0.8985
p=0.675
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

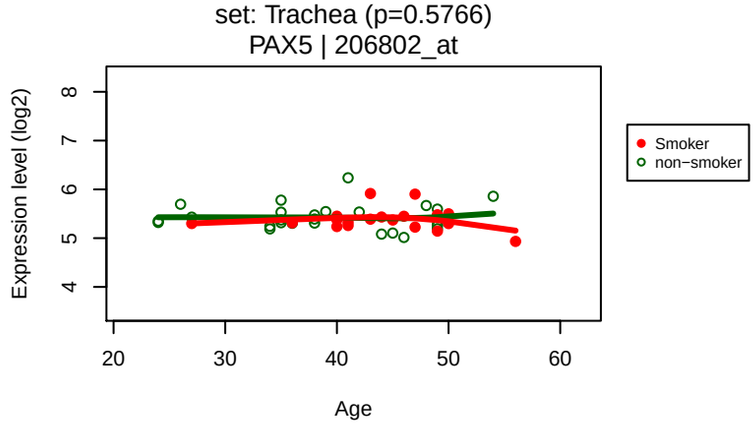

Gene expression ratio in Males
gene: PAX5 | 206802_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0674
p=0.4565
p=0.9285
p=0.7772
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAX5 | 206802_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6533
p=0.136
p=0.1868
p=0.4576
SmallAirway2
LargeAirway2
Trachea
Macrophage2

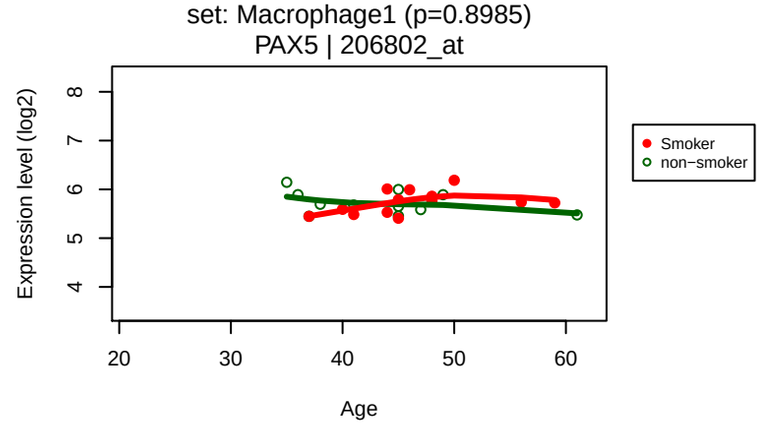

set: SmallAirway1 (p=0.1196)
PAX5 | 206802_at
Expression level (log2)
Age
Smoker
non−smoker

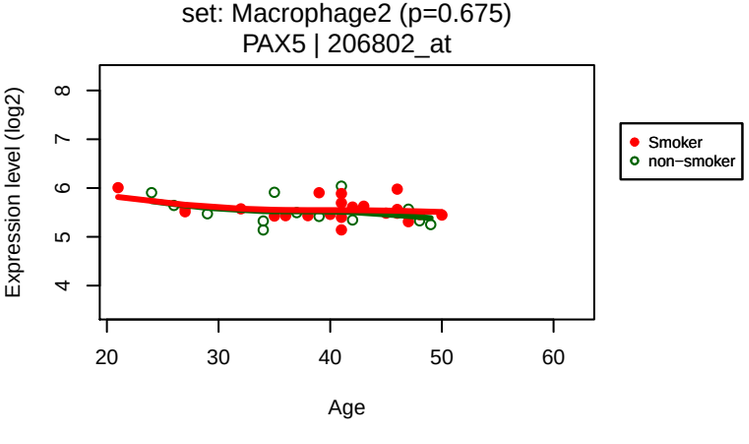

set: SmallAirway2 (p=0.0512)
PAX5 | 206802_at
Expression level (log2)
Age
Smoker
non−smoker

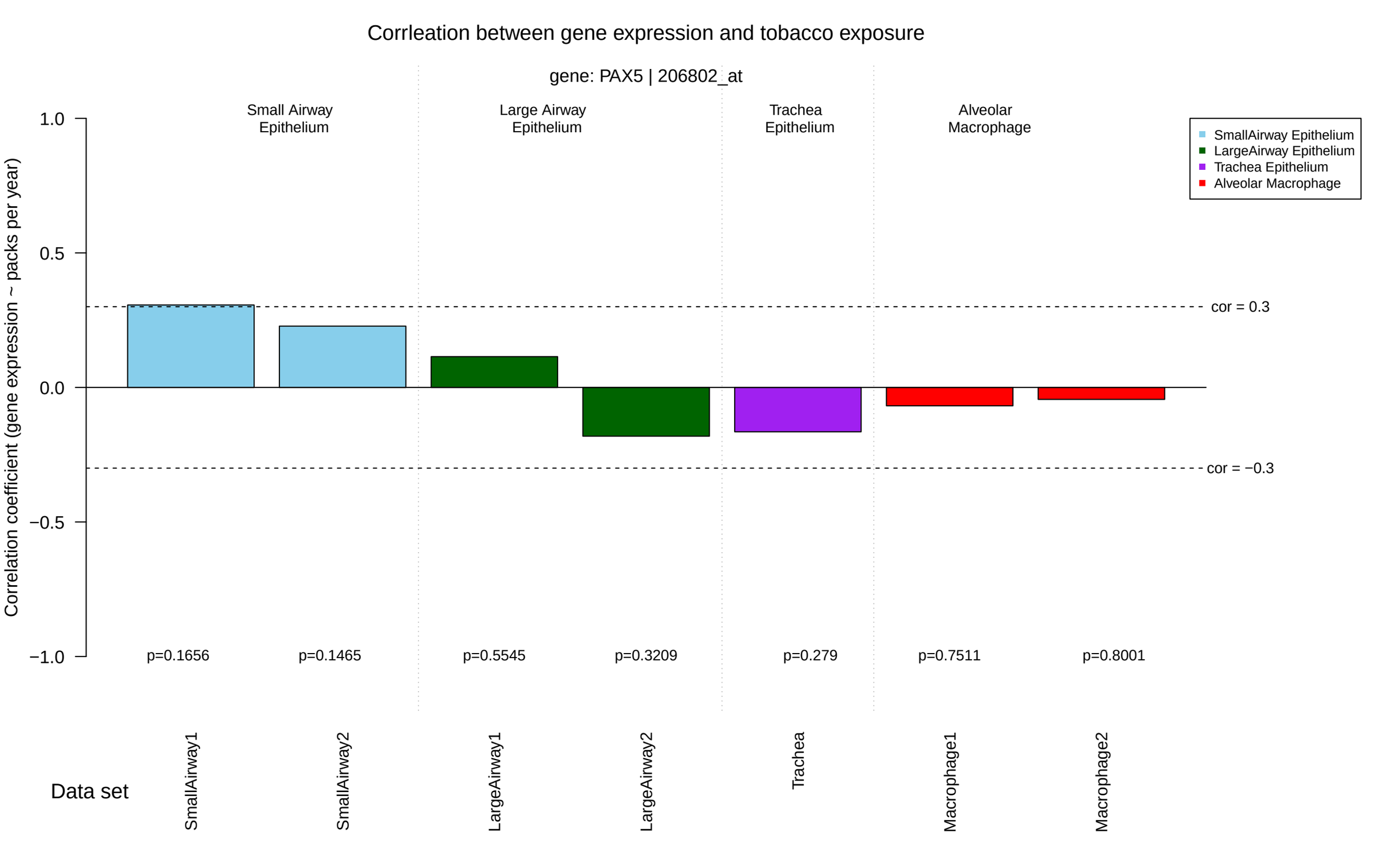

set: LargeAirway1 (p=0.3228)
PAX5 | 206802_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8688)
PAX5 | 206802_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5766)
PAX5 | 206802_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8985)
PAX5 | 206802_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.675)
PAX5 | 206802_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAX5 | 206802_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1656
p=0.1465
p=0.5545
p=0.3209
p=0.279
p=0.7511
p=0.8001
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2